AFTER AN ETALE BASE CHANGE A FINITE TYPE ALGEBRA SPLITS INTO A FINITE PART AND A PART QUASI-FINITE NOWHERE OVER THE POINT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The fibre ring is Noetherian with finitely many isolated closed points. Used at the start of the counted proof (lines 40603-40606). Separates the isolated points of the fibre, which are the pieces that can be made finite.
- The structure of a finite algebra over a field at an isolated point. Used in the counted proof (the splitting of the fibre). Produces the product decomposition of the fibre at the isolated points.

Source reference. Original lemma 00UJ: sources/stacks-algebra/pinned-original-source.tex, source lines 40445–40445. Navigation pointer for \ref{lemma-etale-makes-quasi-finite-finite-one-prime} used inside the counted span (line 40610); the argument that consumes it is retained in the span.

Lemma 0.1. *Let $R \to S$ be a ring map. Let $\mathfrak{p} \subset R$ be a prime. Assume $R \to S$ finite type. Then there exists*

(1) *an étale ring map $R \to R'$,*

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(2) *a prime* $\mathfrak{p}' \subset R'$ *lying over* $\mathfrak{p}$,
(3) *a product decomposition*

$$R' \otimes_R S = A_1 \times \ldots \times A_n \times B$$

with the following properties

(1) *we have* $\kappa(\mathfrak{p}) = \kappa(\mathfrak{p}')$,
(2) *each* A_i *is finite over* R',
(3) *each* A_i *has exactly one prime* $\mathfrak{r}_i$ *lying over* $\mathfrak{p}'$, *and*
(4) $R' \to B$ *not quasi-finite at any prime lying over* $\mathfrak{p}'$.

Proof. Denote $F = S \otimes_R \kappa(\mathfrak{p})$ the fibre ring of S/R at the prime $\mathfrak{p}$. As F is of finite type over $\kappa(\mathfrak{p})$ it is Noetherian and hence $\mathrm{Spec}(F)$ has finitely many isolated closed points. If there are no isolated closed points, i.e., no primes $\mathfrak{q}$ of S over $\mathfrak{p}$ such that S/R is quasi-finite at $\mathfrak{q}$, then the lemma holds. If there exists at least one such prime $\mathfrak{q}$, then we may apply Lemma 00UJ. This gives a diagram

$$\begin{array}{ccccc} S & \longrightarrow & R' \otimes_R S & = & A_1 \times B' \\ \uparrow & & \uparrow & \nearrow & \\ R & \longrightarrow & R' & & \end{array}$$

as in said lemma. Since the residue fields at $\mathfrak{p}$ and $\mathfrak{p}'$ are the same, the fibre rings of S/R and $(A_1 \times B')/R'$ are the same. Hence, by induction on the number of isolated closed points of the fibre we may assume that the lemma holds for $R' \to B'$ and $\mathfrak{p}'$. Thus we get an étale ring map $R' \to R''$, a prime $\mathfrak{p}'' \subset R''$ and a decomposition

$$R'' \otimes_{R'} B' = A_2 \times \ldots \times A_n \times B$$

We omit the verification that the ring map $R \to R''$, the prime $\mathfrak{p}''$ and the resulting decomposition

$$R'' \otimes_R S = (R'' \otimes_{R'} A_1) \times A_2 \times \ldots \times A_n \times B$$

is a solution to the problem posed in the lemma. □